Statistical Inference

Series 14 • Graded Exercises

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Note: This document only contains the graded problems viz Exercise 2 and Exercise 3.

Exercise 2

Let $X_1, \ldots, X_n \overset{iid}{\sim} N_d(\mu, I)$. Construct $100(1-\alpha)\%$ confidence sets/regions for μ by inverting the likelihood ratio test, the Wald's test and the Rao's score test. Is there any connection between the confidence sets? How would you construct confidence sets/regions for μ when $X_1, \ldots, X_n \overset{iid}{\sim} N_d(\mu, \Sigma)$, with $\mu \in \mathbb{R}^d$ and $\Sigma \succeq O$ both unknown?

Solution:

Known covariance. Here $\theta = \mu \in \mathbb{R}^d$. Consider the following

$$\ell(\mu) = -\frac{1}{2}\sum_{i=1}^{n} \|x_i - \mu\|^2 + \text{const} = -\frac{n}{2}\|\bar{X}_n - \mu\|^2 + \text{const}'$$

$$U_n(\mu) = n(\bar{X}_n - \mu), \qquad I_n(\mu) = nI_d, \qquad \hat{\mu} = \bar{X}_n$$

Thus, for testing $H_0 : \mu = \mu_0$ against $H_1 : \mu \neq \mu_0$, we have the following test statistics

$$\text{LRT:} \quad -2\log\Lambda_n = 2\left[\ell(\hat{\mu}) - \ell(\mu_0)\right] = n\|\bar{X}_n - \mu_0\|^2$$

$$\text{Wald:} \quad W_n = (\hat{\mu} - \mu_0)^\top I_n(\hat{\mu})(\hat{\mu} - \mu_0) = n\|\bar{X}_n - \mu_0\|^2$$

$$\text{Score:} \quad S_n = U_n(\mu_0)^\top I_n(\mu_0)^{-1} U_n(\mu_0) = \frac{n^2\|\bar{X}_n - \mu_0\|^2}{n} = n\|\bar{X}_n - \mu_0\|^2$$

Under H_0, $\sqrt{n}(\bar{X}_n - \mu_0) \sim N_d(0, I_d)$, so each statistic is χ^2_d *exactly* (for every n), and each test rejects H_0 if the statistic exceeds $\chi^2_{d,1-\alpha}$. Inverting the three tests, we get the confidence sets

$$C_{\text{LRT}} = C_{\text{Wald}} = C_{\text{Score}} = \left\{\mu \in \mathbb{R}^d : n\|\bar{X}_n - \mu\|^2 \leq \chi^2_{d,1-\alpha}\right\}$$

which is the closed ball centred at $\bar{X}_n$ with radius $\sqrt{\chi^2_{d,1-\alpha}/n}$. Since $n\|\bar{X}_n - \mu\|^2 \sim \chi^2_d$ is a pivot, the coverage probability is exactly $1-\alpha$ for every μ and every n. For $d = 1$ this reduces to $\bar{X}_n \pm z_{1-\alpha/2}/\sqrt{n}$.

Connection. The three confidence sets are *identical*. This happens because $\ell(\mu)$ is exactly quadratic in μ with a constant curvature $-nI_d$ (which does not depend on μ). The Wald and score statistics are the quadratic approximations of the LRT statistic around $\hat{\mu}$ and μ_0 respectively and the information is the same at both points, so the approximation is exact. In general, the three statistics differ only by $o_P(1)$ under H_0.

Unknown covariance. Assume Σ is positive definite and $n > d$. Now Σ is unknown, so the pivot $n(\bar{X}_n - \mu)^\top \Sigma^{-1}(\bar{X}_n - \mu)$ cannot be computed. Consider the following. Since $X_i \overset{iid}{\sim} N_d(\mu, \Sigma)$, we have $\sqrt{n}\,\Sigma^{-1/2}(\bar{X}_n - \mu) \sim N_d(0, I_d)$, and hence

$$n(\bar{X}_n - \mu)^\top \Sigma^{-1}(\bar{X}_n - \mu) \sim \chi^2_d$$

exactly. Let $\hat{\Sigma} = \frac{1}{n}\sum_{i=1}^n (X_i - \bar{X}_n)(X_i - \bar{X}_n)^\top$ be the MLE of Σ. By the law of large numbers, $\hat{\Sigma} \xrightarrow{P} \Sigma$, so $\hat{\Sigma}^{-1} \xrightarrow{P} \Sigma^{-1}$. Thus, by Slutsky's lemma, replacing Σ by $\hat{\Sigma}$ does not change the limit

$$n(\bar{X}_n - \mu)^\top \hat{\Sigma}^{-1}(\bar{X}_n - \mu) \xrightarrow{d} \chi^2_d$$

This is just the Wald statistic with the estimated information $I_n(\hat{\mu}, \hat{\Sigma}) = n\hat{\Sigma}^{-1}$ (for μ). Hence, the $100(1-\alpha)\%$ asymptotic confidence region is

$$\boxed{C = \left\{\mu \in \mathbb{R}^d : n(\bar{X}_n - \mu)^\top \hat{\Sigma}^{-1}(\bar{X}_n - \mu) \le \chi^2_{d,1-\alpha}\right\}}$$

which is an ellipsoid centred at $\bar{X}_n$ (the ball above gets stretched according to $\hat{\Sigma}$, so it is longer in the directions where the data vary more). The LRT and the score test (with Σ replaced by its restricted MLE under H_0) lead to ellipsoids of the same form, which are asymptotically equivalent to C. For $d = 1$ this is $\bar{X}_n \pm z_{1-\alpha/2}\hat{\sigma}/\sqrt{n}$. Also, by the multivariate CLT the same region has asymptotic coverage $1 - \alpha$ even if the X_i are not normal (as long as the covariance matrix is finite and invertible).

Exercise 3

Let $X_1, \ldots, X_n \overset{iid}{\sim}$ Bernoulli(p_1) and $Y_1, \ldots, Y_m \overset{iid}{\sim}$ Bernoulli(p_2) be two independent samples. Construct a $100(1-\alpha)\%$ (asymptotic) confidence interval for $p_1 - p_2$. How would you construct a $100(1-\alpha)\%$ upper confidence interval (of the form $[a, \infty)$) for $p_1 - p_2$?

Solution:

Two-sided interval. The MLEs are $\hat{p}_1 = \bar{X}_n$ and $\hat{p}_2 = \bar{Y}_m$, so that $\hat{\delta} = \hat{p}_1 - \hat{p}_2$ is the MLE of $\delta = p_1 - p_2$ (by invariance). By independence of the two samples

$$E[\hat{\delta}] = \delta, \qquad \mathrm{Var}(\hat{\delta}) = \frac{p_1(1-p_1)}{n} + \frac{p_2(1-p_2)}{m} =: \tau^2$$

Let $n, m \to \infty$ and $p_1, p_2 \in (0,1)$. By the CLT applied to each sample (and independence), $\dfrac{\hat{\delta} - \delta}{\tau} \xrightarrow{d} N(0,1)$. Since $\hat{p}_1 \xrightarrow{P} p_1$ and $\hat{p}_2 \xrightarrow{P} p_2$, the continuous mapping theorem gives $\hat{\tau}/\tau \xrightarrow{P} 1$, where

$$\hat{\tau}^2 = \frac{\hat{p}_1(1-\hat{p}_1)}{n} + \frac{\hat{p}_2(1-\hat{p}_2)}{m}$$

Thus, by Slutsky's lemma, $\dfrac{\hat{\delta} - \delta}{\hat{\tau}} \xrightarrow{d} N(0,1)$, which is an asymptotic pivot (this is the Wald statistic). Hence, the $100(1-\alpha)\%$ asymptotic confidence interval is

$$\boxed{(\hat{p}_1 - \hat{p}_2) \pm z_{1-\alpha/2}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n} + \frac{\hat{p}_2(1-\hat{p}_2)}{m}}}$$

Note that we use the *unpooled* variance estimate since there is no restriction $p_1 = p_2$ (the pooled estimate $\hat{p} = \frac{n\hat{p}_1 + m\hat{p}_2}{n+m}$ is only appropriate for testing $H_0 : p_1 = p_2$).

Upper confidence interval. We invert the one-sided Wald test of $H_0 : \delta = \delta_0$ against $H_{\text{right}} : \delta > \delta_0$, which accepts if $\dfrac{\hat{\delta} - \delta_0}{\hat{\tau}} \leq z_{1-\alpha}$ (we reject for large $\hat{\delta}$ which is evidence of a larger δ). The set of all accepted δ_0 is

$$\{\delta_0 : \hat{\delta} - z_{1-\alpha}\hat{\tau} \leq \delta_0\}$$

Thus, we have the following, using the same asymptotic pivot with the cut-off $z_{1-\alpha}$ instead of $z_{1-\alpha/2}$

$$P\left(\delta \geq \hat{\delta} - z_{1-\alpha}\hat{\tau}\right) = P\left(\frac{\hat{\delta} - \delta}{\hat{\tau}} \leq z_{1-\alpha}\right) \to 1 - \alpha$$

So the $100(1-\alpha)\%$ asymptotic upper confidence interval is

$$\boxed{\left[(\hat{p}_1 - \hat{p}_2) - z_{1-\alpha}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n} + \frac{\hat{p}_2(1-\hat{p}_2)}{m}},\ \infty\right)}$$

Since $\delta \in [-1, 1]$, one may also intersect it with the parameter space, i.e. use $[\max(a, -1), 1]$. As with the two-sided Wald interval, the approximation can be poor if p_1, p_2 are near 0 or 1 (and the interval degenerates to a point if both $\hat{p}_1, \hat{p}_2 \in \{0, 1\}$), in which case one can invert the score test instead.